VINAYAK TRIPATHY

B.Tech Computer Science & Engineering | AI/ML & Software Development
Phone: +91-7008692852 | vinayaktripathy2005@gmail.com | Gunupur, Odisha, India
LinkedIn: linkedin.com/in/vinayaktripathy | GitHub: github.com/akon2005
Portfolio: animated-personal-portfolio-smoky.vercel.app

CAREER OBJECTIVE

Ardently drawn to solving problems through strong fundamentals in Data Structures and Algorithms, backed by hands-on exposure to Machine Learning, Retrieval-Augmented Generation (RAG), and Model Context Protocol (MCP). Eager to leverage my analytical thinking and project experience as a Software Developer, contributing effectively both independently and as part of a team while continuously growing in a fast-paced environment.

EDUCATION

B.Tech, Computer Science & Engineering 2023 – 2027
Gandhi Institute of Engineering and Technology (GIET) University, Gunupur CGPA: 7.6

Intermediate (Class XII), CBSE 2023
Vikash Residential School, Bargarh

Matriculation (Class X), CBSE 2021
Kendriya Vidyalaya, Badmal

TECHNICAL SKILLS

Programming & Data:	Python, SQL, Data Structures and Algorithms
AI / ML:	Machine Learning, Generative AI (Intermediate), LLMs, RAG (Intermediate), MCP (Intermediate), PyTorch, OpenCV, Sentence-Transformers
Web & Vector Search:	FastAPI, React, TailwindCSS, ChromaDB, FAISS

PROJECTS

Retrieval-Augmented Generation Framework for Legal Research and Judicial Knowledge Discovery
Major Project | Python, FastAPI, Sentence-Transformers, ChromaDB/FAISS, React, TailwindCSS

- Built an AI legal research assistant that retrieves similar Indian case precedents and predicts case outcomes from case facts and relevant IPC/BNS sections, returning a confidence score with cited judgments.
- Developed the AI pipeline (embeddings, semantic retrieval, outcome classifier, LLM explanations grounded in retrieved precedents) and the React + TailwindCSS research interface.
- Merged, cleaned and validated the case-law dataset across multiple legal domains; added citation verification against retrieved precedents and low-confidence flags to reduce hallucinated references.

Quantifying Interview Presence with Multimodal AI
Python, OpenCV, PyTorch, Machine Learning

- Built a system that analyzes a candidate's interview video and scores their presence, using OpenCV to track facial expressions, eye contact and posture.
- Trained a PyTorch model to convert frame-by-frame signals into a single usable score, covering the full computer vision pipeline from raw video to output.

INTERNSHIPS

Intern, System Department Jun 2025 – Jul 2025
National Aluminium Company Limited (NALCO), Angul

- Gained exposure to enterprise software and database operations supporting plant workflows.
- Observed workflow automation and testing practices; toured the server room to understand IT infrastructure.

Software Developer Intern Jan 2024 – Feb 2024
Internshala

- Developed scalable software modules in Python and automated data processing tasks.
- Collaborated with the team in an Agile development environment.

CERTIFICATIONS & TRAINING

AI-ML Internship – EduSkills | GenAI & LLMs Internship – EduSkills | Anthropic Courses (Claude, MCP)
Spring Boot – Infosys Springboard | The Joy of Computing using Python – NPTEL
Programming in Java – NPTEL | Adobe Design Fundamentals

SOFT SKILLS

Ownership & Accountability, Strategic Thinking, Team Coordination, Interpersonal Skills, Cross-functional Collaboration, Execution Excellence

ADDITIONAL INFORMATION

Languages: English, Hindi, Odia **Interests:** Driving, reading tech blogs, sports